

Incident response

Condensed Study Notes

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WEKK 5 INCIDENCE RESPONSE

This section introduces the concept of 'incidence response' within the context of WEKK 5. An 'incidence' refers to an event or occurrence that disrupts normal operations. 'Response' refers to the actions taken when such an event happens.

Think of it like a fire alarm in a building. The alarm going off is the 'incidence' – an unexpected event. The 'response' is what people do: evacuate the building, call the fire department, etc. In a technical or organizational context, incidence response involves a planned set of procedures to manage and mitigate the impact of disruptive events.

While the provided content is brief, it establishes that WEKK 5 deals with how to react to or handle specific occurrences that deviate from the expected or desired state.

Introduction

Incident response is a structured and strategic method for dealing with cyber attacks. Its main goals are to reduce the damage caused by an attack, shorten the time it takes to recover, and minimize overall costs.

Think of it like a fire drill for your computer systems. While the fire department (incident response) handles the immediate, technical aspects of putting out the fire, the building management (incident management) oversees the entire process, involving different departments like security, communications, and legal.

Specifically, incident response is a part of a larger concept called incident management. Incident management is the overall way an organization handles cyber attacks, bringing together various teams such as executives, legal, HR, communications, and IT. Incident response focuses on the technical cybersecurity tasks and considerations within that broader management framework.

Types of security incidents

To build effective incident response strategies, it's helpful to understand the relationship between security vulnerabilities, threats, and incidents.

A **vulnerability** is a weakness in an IT or business system. Think of it like an unlocked window in a house.

A **threat** is someone or something that aims to exploit that weakness. This could be a malicious hacker or even an insider. In our house analogy, the threat is the burglar who sees the unlocked window.

An **incident** occurs when an attack succeeds, meaning the threat exploits the vulnerability to access or put enterprise resources at risk. The burglar successfully entering the house through the unlocked window is the incident.

A **data breach** is a specific type of incident where attackers successfully steal sensitive information, like personal details or company secrets. This would be like the burglar taking valuables from the house.

Organizations should proactively fix known vulnerabilities and prepare for common security incidents. These preparedness efforts are crucial because, in cybersecurity, 'an ounce of prevention is worth a pound of cure.'

Common types of security incidents include:

- **Privilege escalation attacks:** Gaining higher access levels than initially intended.
- **Man-in-the-middle attacks:** Intercepting communication between two parties without their knowledge.
- **Malware attacks:** Using malicious software (like viruses or ransomware) to harm systems or steal data.
- **Unauthorized attempts to access systems or data:** Trying to get into systems or data without permission.
- **Advanced persistent threats (APTs):** Sophisticated, long-term attacks often carried out by well-funded groups.
- **Web application attacks:** Targeting vulnerabilities in websites or web-based applications.
- **Insider threats:** Malicious actions taken by individuals within the organization.
- **Phishing attacks:** Tricking individuals into revealing sensitive information, often through deceptive emails or messages.
- **Password attacks:** Trying to guess or crack user passwords to gain unauthorized access.
- **Denial-of-service (DoS) attacks:** Overwhelming a system with traffic to make it unavailable to legitimate users.

How to create an incident response plan

Creating a successful incident response plan involves preparing, reviewing, and testing your strategies *before* a disruptive event occurs. A well-crafted plan acts as a guide, empowering your team to make smart decisions when a crisis hits.

Key components of a robust incident response plan include:

- **Establish a policy:** This is a foundational document that outlines the general, high-level priorities for handling security incidents. Think of it as the 'why' and 'what' of your incident response efforts.
- **Build an incident response team:** The effectiveness of your plan depends on the people involved. Clearly define who is responsible for what tasks and ensure everyone has the necessary training to perform their roles. This is like assembling a skilled emergency response crew.
- **Create playbooks:** If the policy is the 'what', playbooks are the 'how'. These detailed, step-by-step guides provide standardized actions for specific types of security incidents. Playbooks ensure consistency, efficiency, and effective training for your responders. Imagine them as a recipe book for tackling different cyber threats.

- **Create a communication plan:** No incident response can succeed without clear communication. This plan outlines how different stakeholders will be informed and coordinated. Stakeholders can include internal teams (like IT, legal, HR, executives), customers, partners, law enforcement, and the public.

What is an incident response plan

An incident response plan is your organization's essential guide, a comprehensive document that outlines exactly how to handle security incidents. Think of it as a detailed, authoritative map for your incident response team.

This plan guides responders through every stage, from the moment a security incident is first detected, through its assessment and initial sorting (triage), all the way to containing the problem and finally resolving it.

Key elements detailed in the plan include:

- **Who:** Clearly defines who is responsible for which tasks during an incident and provides their contact information.
- **What:** Specifies which threats, exploits, or situations are considered actionable security incidents and outlines the immediate actions to take.
- **How:** Provides specific instructions on how team members should complete their assigned tasks.
- **When:** Determines the specific circumstances under which certain tasks should be performed.

Incident response frameworks: Phases of incident response

Established incident response frameworks, like those from NIST, SANS Institute, ISO, and ISACA, offer high-level guidance for creating an incident response plan. While they may differ slightly, these frameworks generally describe similar phases that an organization follows when responding to a security incident.

Think of these frameworks as a set of blueprints for building a well-structured response. They break down the complex process of handling a cyber attack into manageable stages.

The common phases are:

1. **Preparation/Planning:** This is the foundational stage where an organization gets ready. It involves building an incident response team, creating policies and processes, developing detailed 'playbooks' (step-by-step guides for specific scenarios), and deploying the necessary tools and services to support the response effort.
2. **Detection/Identification:** In this phase, IT monitoring systems are used to detect potential security incidents. Once detected, the incident is evaluated, validated, and triaged to determine its severity and nature.
3. **Containment:** The goal here is to stop the incident from spreading and causing further damage. This involves taking immediate steps to regain control of IT resources that may have been compromised.

4. **Eradication:** This phase focuses on eliminating the threat activity. This includes removing malware, disabling malicious user accounts, and identifying the specific vulnerabilities that attackers exploited so they can be fixed.

5. **Recovery:** Once the threat is eradicated, the focus shifts to restoring normal operations. This also involves mitigating any vulnerabilities that were exploited during the incident to prevent future attacks.

6. **Lessons Learned:** After the incident is resolved and operations are restored, a crucial step is to review what happened. This involves understanding the timeline of events, how the incident occurred, and identifying any security controls, policies, or procedures that didn't function optimally. The aim is to find ways to improve these and update the incident response plan accordingly.

Types of incident response teams

When a security incident occurs, organizations rely on specialized teams to manage the response. The raw content identifies three primary types of these teams:

- **Computer Incident Response Team (CIRT)**
- **Computer Security Incident Response Team (CSIRT)**
- **Computer Emergency Response Team (CERT)**

These teams are dedicated groups responsible for handling cyber attacks and other disruptive events, ensuring a swift and effective response to minimize damage and restore normal operations.

Why is an Incident Response Plan Important

A well-structured incident response plan is crucial for organizations facing cyber threats. It acts as a guide, enabling quick and efficient action when an incident occurs.

Key benefits include:

- **Minimizing the Impact of Incidents:** By following predefined procedures, organizations can reduce downtime, financial losses, and damage to their reputation. Think of it like a fire drill: knowing what to do quickly prevents a small fire from becoming a disaster.
- **Improving Incident Detection:** An effective plan incorporates continuous monitoring and regular reviews of security systems. This helps in detecting threats earlier, allowing for proactive measures to prevent breaches.
- **Streamlining Communication:** A key component is clear communication among all stakeholders, including IT staff, management, and employees. This ensures everyone is informed and working together, reducing confusion and enabling faster recovery.
- **Legal and Regulatory Compliance:** Many industries have strict regulations for data protection and cybersecurity. A documented incident response plan helps organizations demonstrate their commitment to compliance and avoid penalties.
- **Building Resilience:** Developing and maintaining an incident response plan fosters a culture of security awareness. As employees understand potential threats and response steps, the organization becomes more robust against cyberattacks.

